

Esercizio WW

D_t	50	30	20	50	60	30	10	30	40	50
C_t	12									
A_t	80									
h_t	2									

Lot-for-lot

Q_t	50	30	20	50	60	30	10	30	40	50
I_t	0	0	0	0	0	0	0	0	0	0
$C_{\text{set-up}}$	80	80	80	80	80	80	80	80	80	80
C_h	0	0	0	0	0	0	0	0	0	0

$$C_{\text{TOT}} = 80 \times 10 = 800$$

A quantità fissa ($Q=100$)

Q_t	100	0	0	100	100	0	0	0	100	0
I_t	50	20	0	50	90	60	50	20	80	30
C_{setup}	80	0	0	80	80	0	0	0	80	0
C_h	100	40	0	100	180	120	100	40	160	60
C_{TOT}	1220									

WW

$$\textcircled{1} \quad z_1^* = A_1 = 80$$

$$\textcircled{2} \quad J_2^* = 1, 2$$

$$z_2^* = \min \begin{cases} A_1 + h_1 \cdot D_2 = 140 \\ z_1^* + A_2 = 160 \end{cases} \Rightarrow J_2^* = 1$$

$$\textcircled{3} \quad J_3^* = 1, 2, 3$$

$$z_3^* = \min \begin{cases} A_1 + h_1 \cdot (D_2 + D_3) + h_2 \cdot D_3 = 220 \\ z_1^* + A_2 + h_2 \cdot D_3 = 200 \\ z_2^* + A_3 = 220 \end{cases} \Rightarrow J_3^* = 2$$

$$\textcircled{4} \quad J_4^* = 2, 3, 4$$

$$z_4^* = \min \begin{cases} z_1^* + A_2 + h_2 \cdot (D_3 + D_4) + h_3 \cdot D_4 = 400 \\ z_2^* + A_3 + h_3 \cdot D_4 = 320 \\ z_3^* + A_4 = 280 \end{cases} \Rightarrow J_4^* = 4$$

$$\textcircled{5} \quad J_5^* = 4, 5$$

$$z_5^* = \min \begin{cases} (5) \quad z_4^* + A_5 = 360 \\ (4) \quad z_3^* + A_4 + h_4 \cdot D_5 = 400 \end{cases} \Rightarrow J_5^* = 5$$

$$\textcircled{6} \quad J_6^* = 5, 6$$

$$z_6^* = \min \begin{cases} z_4^* + A_5 + h_5 \cdot D_6 = 420 \\ z_5^* + A_6 = 440 \end{cases} \Rightarrow J_6^* = 5$$

$$\begin{aligned}
 & \text{⑦ } J_7^* = 5, 6, 7 \\
 & z_7^* = \min \left\{ \begin{aligned} & z_5^* + A_6 = 440 \\ & z_4^* + A_5 + h_5 \cdot (D_6 + D_7) + h_6 \cdot D_7 = 460 \\ & z_5^* + A_6 + h_6 \cdot D_7 = 460 \Rightarrow J_7^* = 6 \\ & z_6^* + A_7 = 500 \end{aligned} \right.
 \end{aligned}$$

$$\textcircled{8} \quad J_8^* = 6, 7, 8$$

$$z_8^* = \min \begin{cases} z_5^* + A_6 + h_6 \cdot (D_7 + D_8) + h_7 \cdot D_8 = 620 \\ z_6^* + A_7 + h_7 \cdot D_8 = 560 \\ z_7^* + A_8 = 540 \Rightarrow J_8^* = 8 \end{cases}$$

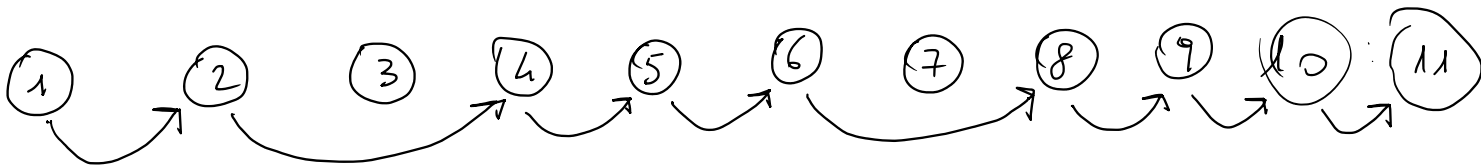
$$\textcircled{9} \quad J_9^* = 8, 9$$

$$z_9^* = \min \begin{cases} z_7^* + A_8 + h_8 \cdot D_9 = 620 \\ z_8^* + A_9 = 620 \Rightarrow J_9^* = 9 \end{cases}$$

$$\textcircled{10} \quad J_{10}^* = 9, 10$$

$$z_{10}^* = \min \begin{cases} z_8^* + A_9 + h_9 \cdot D_{10} = 720 \\ \boxed{z_9^* + A_{10} = 700} \Rightarrow J_{10}^* = 10 \end{cases}$$

Rappresentazione network



$$C_{TOT} = A_1 + A_2 + A_4 + A_5 + A_6 + A_8 + A_9 + A_{10} + h_2 \cdot D_3 + h_6 \cdot D_7 = 700$$

$$Q_1 = D_1 ; Q_2 = D_2 + D_3 ; Q_4 = D_4 ; Q_5 = D_5 ; Q_6 = D_6 + D_7$$

$$Q_8 = D_8 ; Q_9 = D_9 ; Q_{10} = D_{10}$$